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THE MANY-TO-MANY RELATIONSHIP

Reading: Chapter 5

ISE305 – DATABASE DESIGN & PRACTICE

SOME SLIDE MATERIAL FROM WATSON

Objectives

- Learn to model a many-to-many relationship between two entities
- Create a database with a many-to-many relationship
- Write queries for a database with a many-to-many relationship

A Sales Form

Form represents a single sale to a customer, and includes one or more line items

The Expeditioner					
Sale of Goods					
Sale# 123456			Date:		
	Item#	Description	Quantity	Unit price	Total
1					
2					
3					
4					
5					
6					
Grand Total					

A sale might include many items, but a given item might be included in many sales

Example

- A sale includes multiple items
- Items with the same item number might be sold to many customers
- Many-to-many relationship between Items and Sales

The Expeditioner

Sale of Goods

Sale# 123456			Date:		
	Item#	Description	Quantity	Unit price	Total
1					
2					
3					
4					
5					
6					
Grand Total					

itemNo	itemName	itemType	itemColor
1	Pocket knife - Nile	E	Brown
2	Pocket knife - Avon	E	Brown
3	Compass	N	
4	Geopositioning system	N	

Items

ID	itemNo	itemName	itemType	itemColor
1	1	Pocket knife - Nile	E	Brown
2	2	Pocket knife - Avon	E	Brown
3	3	Compass	N	
4	4	Geopositioning system	N	
5	5	Map measure	N	
6	6	Hat - Polar Explorer	C	Red
7	7	Hat - Polar Explorer	C	White
8	8	Boots - snake proof	C	Green
9	9	Boots - snake proof	C	Black
10	10	Safari chair	F	Khaki
11	11	Hammock	F	Khaki
12	12	Tent - 8 person	F	Khaki
13	13	Tent - 2 person	F	Khaki
14	14	Safari cooking kit	E	
15	15	Pith helmet	C	Khaki
16	16	Pith helmet	C	White
17	17	Map case	N	Brown
18	18	Sextant	N	
19	19	Stetson	C	Black
20	20	Stetson	C	Brown

Sales

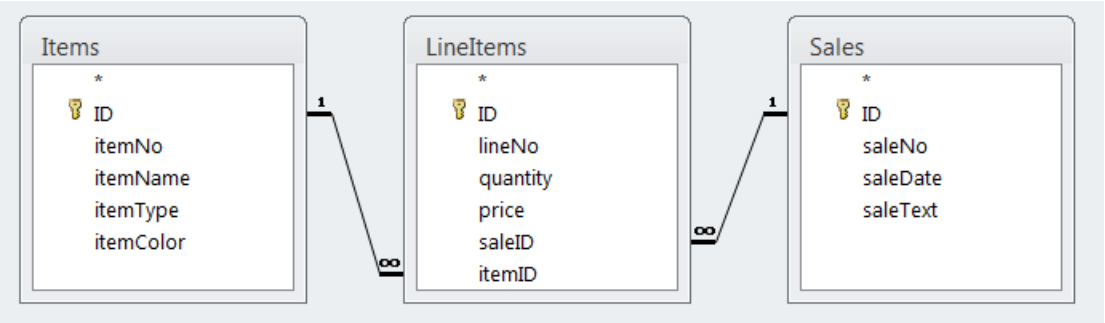
- LineItems table refers to both the Sales table and the Items table

ID	saleNo	saleDate	saleText
1	1	1/15/2003	Scruffy Australian - called himself Bruce
2	2	1/15/2003	Man. Rather fond of hats.
3	3	1/15/2003	Woman. Planning to row Atlantic - lengthwise
4	4	1/15/2003	Man. Trip to New York - Thinks NY is a jungle!
5	5	1/16/2003	Expedition leader for African safari

Many-to-Many Relationships

- Many-to-many relationships cannot be represented directly in a relational DB
- You refine the relationship to include an associative (or "in-between") entity with only one-to-many relationships

Workbench will do this automatically when you drag a m:n relationship between 2 entities



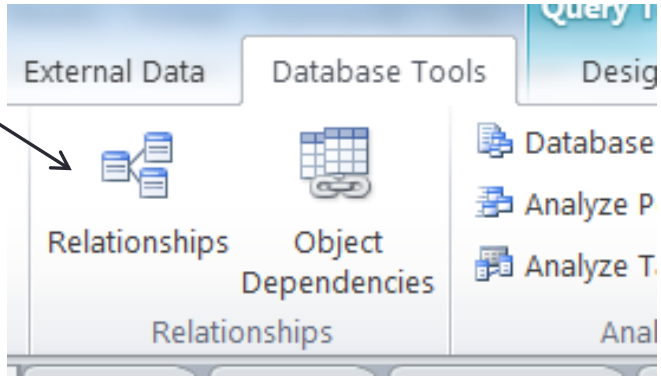
Why a Third Entity?

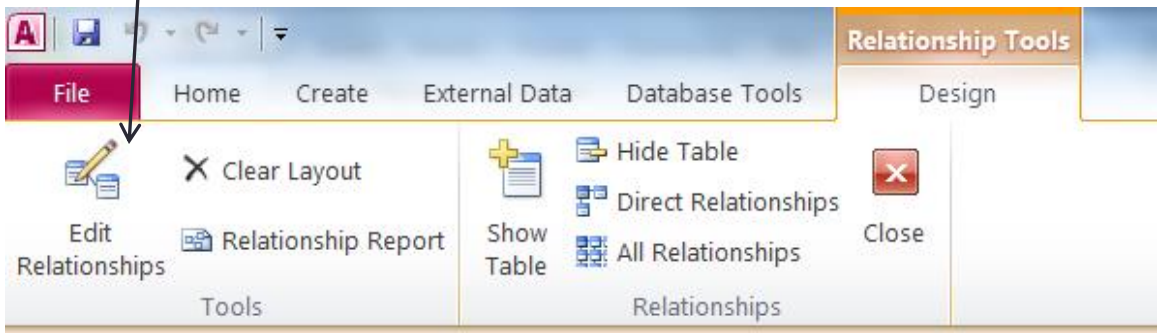
- Store data about the relationship
- Usually represents a real-world entity (e.g., line items)
- Can have its own attributes
- Think of an n:m as two 1:n relationships

E.g., A line item would include the quantity (of a given item) and the sale price of that item.

Creating Relationships in Access

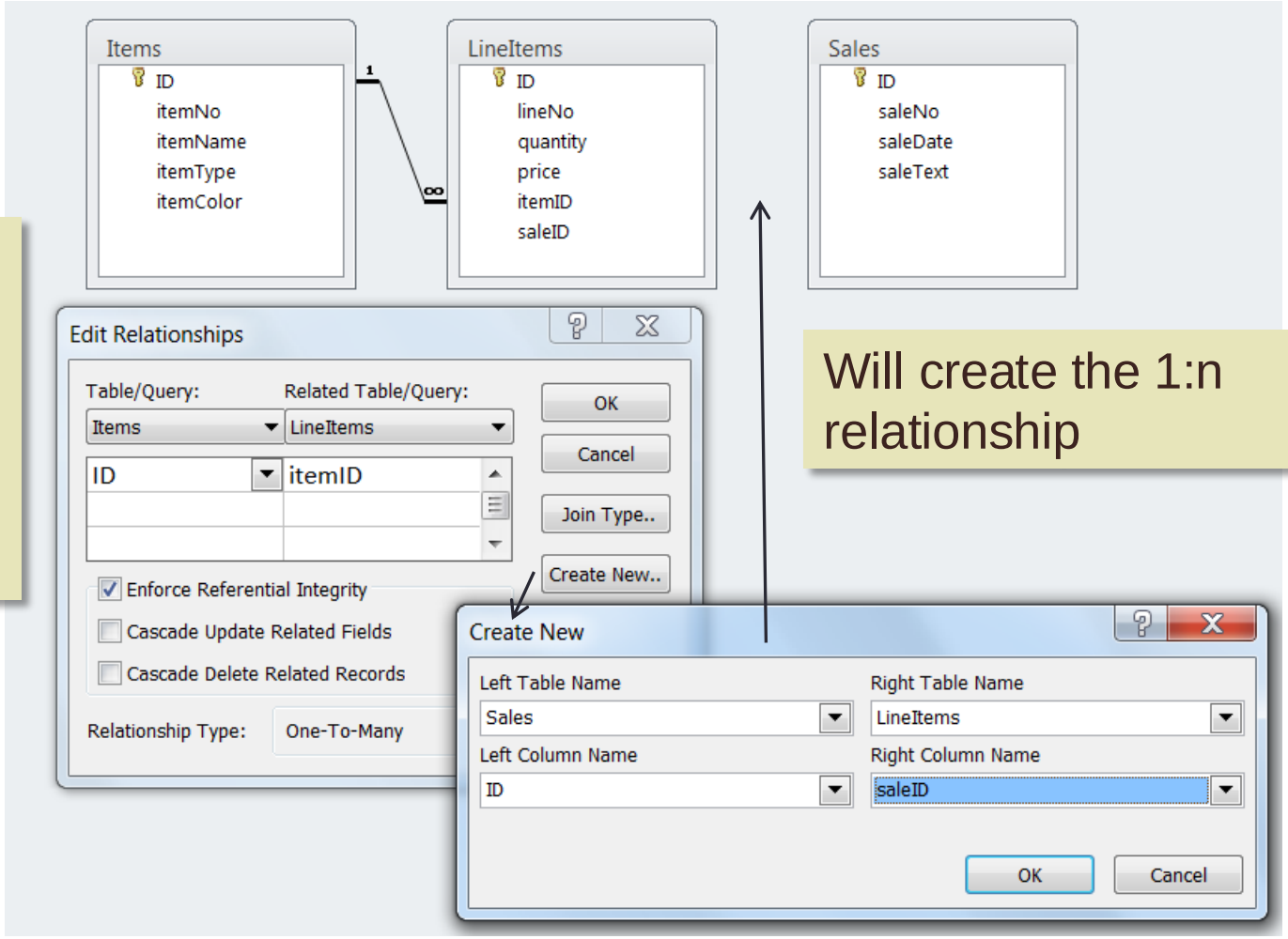
- To edit relationships in your DB
  - Click on Database Tools / Relationships
  - Click on Edit Relationships





Creating Relationships in Access

You can also drag the primary key to the foreign key in the second table



Will create the 1:n relationship

Associative Table

- Has two foreign keys - one for each of the entities in the n:m relationship
- Contains other attributes consistent with the table

Textbook uses natural keys and a composite primary key for LineItems – we use non-natural keys

Queries of the many-to-many relationship usually requires a join of more than 2 tables

LineItems

ID	lineNo	quantity	price	saleID	itemID	C
1	1	1	$4.50	1	2	
2	1	1	$25.00	2	6	
3	2	1	$20.00	2	16	
4	3	1	$25.00	2	19	
5	4	1	$2.25	2	2	

A Three Table Join

- List the names of the three tables after **FROM**
- Specify two matching conditions with the associative table in both join conditions

```
SELECT * FROM Sales, LineItems, Items
  WHERE Sales.ID = LineItems.saleID
    AND Items.ID = LineItems.itemID;
```

LineItems | ThreeTableJoin

Sales.ID	saleNo	saleDate	saleText	LineItems.ID	lineNo	quantity	price
1	1	1/15/2003	Scruffy Australian - called himself Bruce	1	1	1	$4
2	2	1/15/2003	Man. Rather fond of hats.	2	1	1	$25
2	2	1/15/2003	Man. Rather fond of hats	3	2	1	$20

Example - Three Table Join

SELECT saleNo, saleDate, itemName, itemType, quantity, price
FROM Sales INNER JOIN (Items INNER JOIN LineItems ON Items.ID = LineItems.itemID) ON Sales.ID = LineItems.saleID
ORDER BY Sales.saleNo;

saleNo	saleDate	itemName	itemType	quantity	price
1	1/15/2003	Pocket knife - Avon	E	1	$4.50
2	1/15/2003	Hat - Polar Explorer	C	1	$25.00
2	1/15/2003	Pith helmet	C	1	$20.00
2	1/15/2003	Stetson	C	1	$25.00
2	1/15/2003	Pocket knife - Avon	E	1	$2.25
3	1/15/2003	Geopositioning system	N	1	$500.00
3	1/15/2003	Pocket knife - Avon	E	1	$2.25
4	1/15/2003	Tent - 2 person	F	1	$60.00
4	1/15/2003	Boots - snake proof	C	1	$65.00
4	1/15/2003	Safari cooking kit	E	1	$75.00
4	1/15/2003	Compass	N	1	$10.00
4	1/15/2003	Pocket knife - Avon	E	1	$2.25
4	1/15/2003	Geopositioning system	N	1	$500.00
5	1/16/2003	Pocket knife - Avon	E	1	$0.00
5	1/16/2003	Safari chair	F	50	$36.00
5	1/16/2003	Hammock	F	50	$40.50
5	1/16/2003	Tent - 8 person	F	8	$153.00
5	1/16/2003	Tent - 2 person	F	1	$60.00

Example

- List the dates, names of items, quantity, and value of items

```
SELECT saleDate, itemName, quantity, price,
  quantity*LineItems.price AS total
FROM Sales, LineItems, Items
WHERE LineItems.saleID = Sales.ID
  AND  LineItems.itemID = Items.ID;
```

DaySummaryQuery

saleDate	itemName	quantity	price	total
1/15/2003	ocket knife - Avon	1	$4.50	$4.50
1/15/2003	Hat - Polar Explorer	1	$25.00	$25.00
1/15/2003	Pith helmet	1	$20.00	$20.00
1/15/2003	Stetson	1	$25.00	$25.00
1/15/2003	Pocket knife - Avon	1	$2.25	$2.25
1/15/2003	Geopositioning system	1	$500.00	$500.00
1/15/2003	Pocket knife - Avon	1	$2.25	$2.25
1/15/2003	Geopositioning system	1	$500.00	$500.00

Example 2

- List the names of items, quantity, and value of items sold on January 16, 2003

Note the date literal syntax is different from text

```
SELECT saleDate, itemName, quantity,
  price, quantity*LineItems.price AS total
FROM Sales, LineItems, Items
WHERE LineItems.saleID = Sales.ID
      AND  LineItems.itemID = Items.ID
  AND saleDate = #1/16/2003#;
```

SummaryForDateQuery | Sales

saleDate	itemName	quantity	price	total
1/16/2003	Safari chair	50	$36.00	$1,800.00
1/16/2003	Hammock	50	$40.50	$2,025.00
1/16/2003	Tent - 8 person	8	$153.00	$1,224.00
1/16/2003	Tent - 2 person	1	$60.00	$60.00
1/16/2003	Pocket knife - A	1	$0.00	$0.00

Are We on Track Setup

- Set up a new database using the data from Chapter 5 in the Watson text
- Import from Blackboard
  - Items.xlsx
  - Sales.xlsx
  - ExportLineItems.xlsx (store as LineItems table)
- Add foreign keys to LineItems table and establish relationships

You can set the foreign keys in LineItems using values real values of the primary keys

Are We on Track?

- Using Design View in Access, write a query to:
  - Compute the total value of sales of each item by date
  - Primary sort by date
  - Include a count of the items sold on each date

itemName	saleDate	Number Sold	Total Sales
Boots - snake proof	1/15/2003	1	$65.00
Compass	1/15/2003	1	$10.00
Geopositioning system	1/15/2003	2	$1,000.00
Hat - Polar Explorer	1/15/2003	1	$25.00
Pith helmet	1/15/2003	1	$20.00
Pocket knife - Avon	1/15/2003	4	$11.25
Safari cooking kit	1/15/2003	1	$75.00
Stetson	1/15/2003	1	$25.00
Tent - 2 person	1/15/2003	1	$60.00
Hammock	1/16/2003	1	$40.50
Pocket knife - Avon	1/16/2003	1	$0.00
Safari chair	1/16/2003	1	$36.00
Tent - 2 person	1/16/2003	1	$60.00
Tent - 8 person	1/16/2003	1	$153.00

Homework

- Using the Awards DB (multi-table), develop a query to:
  - Show number of awards, total budget by sponsor
  - Sorted by total budget

JoinQuery | Awards | Relationships

sponsorName	totalBudget	numberAwards
National Science Foundation	$101,197,276.00	251
US Department of Energy	$71,826,927.25	33
Empire State Development Corp	$60,914,498.00	4
National Institute of Allergy & Infectious Disease	$55,867,955.47	29
National Institute of General Medical Sciences	$41,472,116.33	54
Dormitory Authority of the State of New York	$36,542,000.00	2
National Cancer Institute	$30,104,236.81	34
National Inst of Neurological Disorders & Stroke	$19,983,384.67	22

Homework

- Assume that awards are allowed to have multiple PIs.
- Update the SBU_Awards DB accordingly
- Add additional PIs to some of the awards
- Hint:
  - Create a new PIs table with a handful of principle investigators
  - When you create the “in-between” table, add an additional attribute for Percentage (the percentage of the contract for the given PI

EXISTS

itemName	itemColor
Hat - Polar Explorer	Red
Boots - snake proof	Black
Pith helmet	White
Stetson	Black

- Existential qualifier
- Returns true or false
- Returns true if the table contains at least one row satisfying the specified condition
- Report all clothing items (type “C”) for which a sale is recorded

```
SELECT itemName, itemColor FROM Items
WHERE itemtype = 'C'
  AND EXISTS
     (SELECT * FROM LineItems
      WHERE LineItems.itemID = Items.ID);
```

The sub-query is evaluated for each row of Items. Resulting row is only added to result table if EXISTS evaluates to TRUE

NOT EXISTS

- Returns true if the table contains no rows satisfying the specified condition
- Example: report all clothing items (type “C”) that have not been sold

itemName	itemColor
Hat - Polar Explorer	White
Boots - snake proof	Green
Pith helmet	Khaki
Stetson	Brown

```
SELECT itemname, itemcolor FROM Items
WHERE itemtype = 'C'
  AND NOT EXISTS
   (SELECT * FROM lineitem
    WHERE item.itemno = lineitem.itemno);
```

Divide

- The universal quantifier
  - ForAll (uses the symbol ∀)
  - Meaning – something is true for everything or every relevant thing
- Not directly mapped into SQL
- Implement using NOT EXISTS
  - Find all items that have appeared in all sales becomes
  - Find items such that there does not exist a sale in which this item does not appear

Not a very natural expression of the condition

Divide Example

- Find the items that have appeared in all sales

```
SELECT itemNo, itemName FROM Items
  WHERE NOT EXISTS
    (SELECT * FROM Sales
      WHERE NOT EXISTS
        (SELECT * FROM LineItems
  WHERE LineItems.itemID = Items.ID
    AND LineItems.saleID = Sales.ID));
```

itemNo	itemName
2	Pocket knife - Avon

Divide Reformulation

- Find the items that have appeared in all sales can be rephrased as
  - Find all the items for which the number of sales that include this item is equal to the total number of sales.

```
SELECT Items.ID, Items.itemName
FROM Items, LineItems
WHERE Items.ID = LineItems.itemID
GROUP BY Items.ID, Items.itemName
HAVING
  COUNT(LineItems.saleID) = (SELECT COUNT(Sales.ID) FROM Sales);
```

Set Operations

- Useful in combining values derived from two or more queries
- UNION
  - Equivalent to OR
- INTERSECT
  - Equivalent to AND

UNION Example

- List all items that were sold on January 16, 2003, or are brown

```
SELECT itemName FROM Items, LineItems, Sales
WHERE Items.ID = LineItems.itemID
  AND LineItems.saleID = Sales.ID
  AND Sales.saleDate = #2003-01-16#
UNION
  SELECT itemName FROM Items
    WHERE itemColor = 'Brown';
```

itemName
Hammock
Map case
Pocket knife - Avon
Pocket knife - Nile
Safari chair
Stetson
Tent - 2 person
Tent - 8 person

INTERSECT

- List all items that were sold on January 16, 2003, and are brown.

```
SELECT itemName FROM Items, LineItems, Sales
WHERE Items.ID = LineItems.itemID
  AND LineItems.saleID = Sales.ID
  AND Sales.saleDate = #2003-01-16#
INTERSECT
  SELECT itemName FROM Items
    WHERE itemColor = 'Brown';
```

INTERSECT not supported by some DBMSs (e.g., Access)

Are We on Track?

- Report all items of ‘F’ that have been sold.

itemName	itemType
Safari chair	F
Hammock	F
Tent - 8 person	F
Tent - 2 person	F

Have You Satisfied the Objectives

- Learn to model a many-to-many relationship between two entities
- Create a database with a many-to-many relationship
- Write queries for a database with a many-to-many relationship